

Pharmacy Ordering System (InfoPharma Ltd) Software

Summary

This is the Requirements Specification and High-Level Design Document, a proposal to the Founder and CEO of InfoPharma Ltd, by Zaina Meraj, Project Manager at Softwave.

Authors

Zaina Meraj, Yaseen Barlas, Sukhmanvir Atwal, Jagdish Vaghela, Viktor Vasilchin, Rizwan Hussen, Leo Haghighi

Distribution

The Founder and CEO of InfoPharma Ltd, the Development Team at Softwave.

Contents

1 Preface.....	3
1.1 Purpose and scope of the document (what it is and what is NOT)	3
1.2 Intended audience and Reading Suggestions	3
1.3 History of the document	3
2 Introduction.....	3
3 Glossary	3
4 User requirements	4
5 System requirements	5
5.1 Purpose and scope of the system	5
5.2 System models	5
5.2.1 UML use-case models	6
6 Anticipated Evolution of System Requirements	6
6.1 Validation	8
6.2 Loyalty Cards	8
7 Dental Pro High -Level Design	8
7.1 Dental-Core	8
7.2 Dental-Patients	8
7.3 Dental-Inventory	8
7.4 Software architecture	8
7.4. Design diagram of subsystem Dental-Core	10
7.5 Source code (Java) for the interfaces of Dental-Core	11
7.6 Testing the communications between subsystems	13
8 Appendices	14
8.1 Initial Statement of requirements by the client	14
Dental Pro: Dental Practice Management Software	14
8.2 Hardware (minimal and optimal configuration)	16
8.2.1 Minimal hardware configuration	16
8.2.2 Optimal hardware configuration	16
9 Index	16

1 Preface

1.1 Purpose and scope of the document (what it is and what is NOT)

This document is a Requirements Specification and High-Level Design document for the InfoPharma Ordering System (IPOS). It describes the functional requirements of the IPOS-CA (Client Application) subsystem and presents the high-level design of the overall IPOS software, which consists of three interacting subsystems: IPOS-SA, IPOS-CA, and IPOS-PU.

The document explains what the IPOS-CA subsystem shall do and how it is expected to interact with the other IPOS subsystems. It does not include details of implementation such as user interface layouts, database schemas, or low-level program logic.

The document is provided in response to the initial statement of requirements issued by InfoPharma Ltd, the client for this project. UML models and high-level interface descriptions are used to convey the expected behaviour of the IPOS-CA subsystem and the mechanisms of interaction between the subsystems to system designers and software developers.

1.2 Intended audience and Reading Suggestions

The intended audience of this document includes representatives of InfoPharma Ltd, as well as system analysts, architects, designers, and software developers involved in the development of the IPOS software.

Some sections of the document are primarily intended for the development team. Readers without a technical background are encouraged to focus on the introductory sections and overview descriptions.

1.3 History of the document

This is a new document, v. 1.0.

2 Introduction

The InfoPharma Ordering System (IPOS) is intended to support the distribution and retail of pharmaceutical products by automating key business processes for InfoPharma Ltd and its pharmacy customers.

IPOS aims to improve the efficiency, accuracy, and reliability of current business operations by providing computer-based support for merchant ordering, customer account management, local stock control, sales processing, and reporting. The system is designed to reduce reliance on manual processes and to minimise errors related to ordering, stock shortages, and customer debt handling.

The IPOS software is structured around three interconnected subsystems. IPOS-SA supports InfoPharma Ltd's internal operations, including catalogue management, merchant accounts, order processing, and invoicing. IPOS-CA is a client application used by pharmacy shops to manage local retail operations such as placing orders, managing customers, handling payments, and maintaining local stock. IPOS-PU is an online platform that enables pharmacy merchants to sell pharmaceutical products directly to members of the public.

This document focuses specifically on IPOS-CA, the subsystem responsible for supporting day-to-day operations within pharmacy shops. IPOS-CA interacts with the other subsystems as required to fulfil its functionality, but those subsystems are specified in separate documents.

3 Glossary

Actor	A human user or an external system that interacts with IPOS, such as a Pharmacist, Manager, or another IPOS subsystem.
Customer	An individual purchasing pharmaceutical products from a pharmacy using IPOS-CA. Customers may be account holders or occasional customers.
InfoPharma Ltd	The pharmaceutical distribution company for which the IPOS system is being developed.
IPOS	InfoPharma Ordering System.
IPOS-CA	Client Application subsystem of IPOS, used by pharmacy shops to manage orders, customers, stock, sales, and reports.
IPOS-SA	Server Application subsystem of IPOS, operated by InfoPharma Ltd, responsible for catalogue management, merchant accounts, and order processing.
IPOS-PU	Public-facing subsystem of IPOS that enables online sales of pharmaceutical products.
Subsystem	A major functional component of IPOS that provides a distinct set of services and interacts with other subsystems via defined interfaces.

UML	Unified Modelling Language, used to model system requirements and high-level design.
Use Case	A description of a basic interaction between an actor and the system to achieve a specific goal.

4 User Requirements

This section is left intentionally blank

4 System Requirements

5.1 Purpose and scope of the system

This section provides a more detailed specification of required functionality (known as system requirements) for IPOS-CA.

The intended audience is the system architects, designers, and software developers for the implementation of IPOS-CA. Since we use UML, which requires a degree of specific expert knowledge, we do not expect this section to be immediately useful to the InfoPharma staff but would like to encourage them to review and comment if possible

5.2 System Models

5.2.1 UML use-case models

5.2.1.1 Use case diagram

The use case diagram for IPOS-CA subsystem is shown below in Figure 1. In this diagram the other subsystems of IPOS-CA are treated as “external systems”, with which IPOS-CA may interact.

5.2.1.1 Use case specifications (simplified)

ID: CA-01	Use Case: Log In
Brief Description: Allows the pharmacist to authenticate themselves to gain access into the IPOS-CA subsystem. The system verifies the log in credentials of the staff before allowing access into the subsystem.	
Primary Actors:	

User
Secondary Actors: None
Preconditions: 1. System must be functional. 2. User must have an existing account with valid log in credentials.
Postconditions: 1. User is successfully logged into the system.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. InvalidCredentials 2. LockedAccount

ID: CA-02	Use Case: Log Out
Brief Description: Allows a user who is currently logged into the system to successfully logout of the system to prevent any unauthorised access into the pharmacy system.	
Primary Actors: User	
Secondary Actors: None	
Preconditions: 1. System is fully operational. 2. User is logged into the system using valid credentials.	

Postconditions: 1. The user must not have access in the pharmacy system at all after logging out.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. SessionTimedOut

ID: CA-03	Use Case: Create User Account
Brief Description: Allows for a customer account to be created in the pharmacy system, which takes the customer's first name, surname, DOB, email address, home address and phone number. This is done to store customer details on the pharmacy system for future transactions.	
Primary Actors: Admin	
Secondary Actors: None	
Preconditions: 1. System is fully operational.	

- Admin is logged into the system. - Inputs of valid first name, surname, DOB, email address, home address and phone number into required fields.
Postconditions: Customer's data is recorded and stored onto the pharmacy system successfully.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: - InvalidInput (e.g., user enters '#' within the surname field). - DuplicateCustomer

ID: CA-04	Use Case: Remove User Account
Brief Description: Removes the existing user account, preventing access into the IPOS-CA system.	
Primary Actors: Admin	
Secondary Actors: None	
Preconditions: - System is fully operational. - User account exists in the IPOS-CA system. - Admin is aware of which account to be removed.	
Postconditions:	

1. Account has been successfully removed from the system.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. UserNotFound

ID: CA-05	Use Case: Update Account Holder Details
Brief Description:	
Primary Actors: Pharmacist	

Updates the existing details which are stored of an existing account holder. This is done to ensure account holder details remain accurate.	
Secondary Actors:	
None	
Preconditions:	
1. System is fully operational. 2. There is an existing account holder. 3. The pharmacist is logged into the system successfully using their credentials. 4. The pharmacist aware of the details which need to be updated.	
Postconditions:	
1. Account holder details are updated successfully.	
Extension Use Case:	
None	
Inclusion Use Case:	
None	
Alternative Flows:	
1. AccountHolderNotExisting 2. InvalidDetailsToUpdate	

ID: CA-06	Use Case: View Account Status
Brief Description:	
Displays the status of the account holder, which includes the balance and checking statements.	
Primary Actors:	

Pharmacist
Secondary Actors: None
Preconditions: 1. System is fully operational. 2. There is an existing account holder. 3. The pharmacist will access the account holder status.
Postconditions: 1. Account status is displayed to the user
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. AccountHolderNotExisting 2. AccountHolderSuspended

ID: CA-07	Use Case: Generate Monthly Statements
Brief Description: A monthly statement will be generated which contains all transactions which have been made within that particular month, helping to support the customer in reviewing their past transactions.	
Primary Actors: Pharmacist	

Secondary Actors: None
Preconditions: 1. System is fully operational. 2. There is an existing account holder. 3. Monthly transactions are present within the system.
Postconditions: 1. Monthly statements are generated successfully.
Extension Use Case: None
Inclusion Use Case: • Generate 1st Reminder • Generate 2nd Reminder
Alternative Flows: 1. NoTransactionsMade

ID: CA-08	Use Case: Assign User Role
Brief Description: A role will be assigned to the user account, which demonstrates how much of the system can be accessed within the pharmacy's subsystem.	
Primary Actors: Admin	
Secondary Actors: None	

Preconditions: 1. System is fully operational. 2. A user account exists on the system. 3. Admin has the correct login credentials to access the system. 4. Admin is aware of the appropriate role to assign to each user.
Postconditions: 1. Admin has successfully assigned a role to a user account.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. InvalidRoleEntered 2. NoUserPresent

ID: CA-09	Use Case: Update Account Status
Brief Description: Updates the status of the account holder, such as the account being active or suspended.	
Primary Actors: IPOS-CA	
Secondary Actors: None	
Preconditions:	

1. System is fully operational. 2. Account holder exists in the system. 3. The subsystem is aware of which account to update the status on.
Postconditions: 1. Account holder's status is updated.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. AccountNotFound 2. InvalidStatusProvided

ID: CA-10	Use Case: Record Account Payment
Brief Description: Any payments made by the account holder is recorded, updating the outstanding balance.	
Primary Actors: Pharmacist	
Secondary Actors: None	
Preconditions: 1. System is fully operational. 2. Account holder has an outstanding balance which has been already paid.	

Postconditions: 1. Payment is recorded. 2. Outstanding balance is updated.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. PaymentDeclined

ID: CA-11	Use Case: Generate 1 st Reminder
Brief Description: A first reminder is generated for accounts which have overdue payments.	
Primary Actors: IPOS-CA	
Secondary Actors: None	
Preconditions: 1. System is fully operational. 2. Account holder exists in the system. 3. Account holder has a payment which is overdue.	
Postconditions:	

1. Reminder will be generated.
Extension Use Case: 1. Generate 2 nd Reminder
Inclusion Use Case: None
Alternative Flows: 1. FailToSendReminder

ID: CA-12	Use Case: Generate 2 nd Reminder
Brief Description: A second reminder will be generated if the payment is still overdue after the first reminder.	
Primary Actors: IPOS-CA	
Secondary Actors: None	
Preconditions: 1. System is fully operational. 2. First reminder is already sent to the account holder. 3. Second reminder is sent to the account holder.	
Postconditions:	

1. Second reminder has been generated and sent to the account holder.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. FailedToSendReminder

ID: CA-13	Use Case: Record Sale
Brief Description: A record is made of every sale transaction which is completed within the pharmacy system.	
Primary Actors: Pharmacist	
Secondary Actors: None	
Preconditions: 1. System is fully operational. 2. A sale is completed on the pharmacy system.	
Postconditions: 1. Sale is recorded.	

Extension Use Case:

None

Inclusion Use Case:

1. Generate Receipt/Invoice
2. Accept Payment

Alternative Flows:

1. SaleRecordingFailed

ID: CA-14**Use Case:** Accept Payment**Brief Description:**

A payment will be accepted for a sale or any account transactions which have been made.

Primary Actors:

Pharmacist

Secondary Actors:

None

Preconditions:

1. System is fully operational.
2. A payment will be required for any pharmaceutical product which is being purchased.

Postconditions:

1. Payment will be accepted.

Extension Use Case:

None

Inclusion Use Case:

None

Alternative Flows:

1. PaymentFailed

ID: CA-15

Use Case: Process Cash Payment

Brief Description:

Processes payments which are made using cash.

Primary Actors:

One off Customer

Secondary Actors:

None

Preconditions:

1. System is fully operational.
2. Customer picks a pharmaceutical product they'd like to purchase
3. Method of payment selected by the customer is cash.

Postconditions:

1. Cash payment is processed.

Extension Use Case:

None

Inclusion Use Case:

None

Alternative Flows:

1. InsufficientCash

ID: CA-16**Use Case:** Process Card Payment**Brief Description:**

Processes payments which are made using a card.

Primary Actors:

1. Customer (One off Customer or Account Holder Customer)

Secondary Actors:

1. IPOS-PU

Preconditions:

1. System is fully operational.
2. Customer chooses a pharmaceutical product they'd like to purchase.
3. Method of payment chosen by the customer is by card.

Postconditions:

1. Card payment is processed.

Extension Use Case:

None

Inclusion Use Case:

None

Alternative Flows:

1. InsufficientFundsInCard

ID: CA-17

Use Case: Generate Receipt/Invoice

Brief Description:

A receipt/invoice is generated when a successful transaction is completed.

Primary Actors:

IPOS-CA

Secondary Actors:

None

Preconditions:

1. System is fully operational.
2. Payment is successfully completed for the pharmaceutical product the customer wanted to purchase.

Postconditions:

1. Receipt/Invoice is generated by the subsystem (IPOS-CA).

Extension Use Case:

None

Inclusion Use Case:

None
Alternative Flows: 1. PrintError 2. PaymentFailed

ID: CA-18	Use Case: Maintain Stock Records
Brief Description: Provides accurate records for the level of stock of pharmaceutical products.	
Primary Actors: Pharmacist	
Secondary Actors: IPOS-PU	
Preconditions: 1. Systems are fully functional. 2. There is existing inventory of pharmaceutical products. 3. Pharmacist and IPOS-PU subsystem is aware of which product's stock levels need updating.	
Postconditions: 1. Stock records are updated.	
Extension Use Case: None	
Inclusion Use Case: None	

Alternative Flows:

1. NoStock
2. InventoryUpdateFailed

ID: CA-19**Use Case:** View Stock Availability**Brief Description:**

Allows the merchant to view current stock levels for all products, including quantities available and basic product details.

Primary Actors:

Pharmacist

Secondary Actors:

IPOS-PU

Preconditions:

1. Systems are fully functional.
2. Pharmacist is authenticated.
3. Product and stock data exists within the system.

Postconditions:

1. Current stock availability is displayed to Pharmacist

Extension Use Case:

None

Inclusion Use Case:

1. Order More Stock
2. Check If Stock Low

Alternative Flows:

None

ID: CA-20	Use Case: Check If Stock low
Brief Description: Automatically checks stock level against the predefined thresholds and identifies products with low stock	
Primary Actors: Pharmacist	
Secondary Actors: None	
Preconditions: 1. System is fully functional. 2. Stock availability data exists. 3. Low-stock thresholds are defined.	
Postconditions: 1. Low-stock items are highlights/flagged	
Extension Use Case: None	
Inclusion Use Case: None	
Alternative Flows:	

1. DataNotRetrieved

ID: CA-21	Use Case: Configure VAT
Brief Description: Allows the Pharmacist to configure VAT rates that are applied to products and transactions.	
Primary Actors: Pharmacist	
Secondary Actors: None	
Preconditions: 1. System is fully functional. 2. Pharmacist is logged into the system using their credentials. 3. Pharmacist is aware of what VAT to configure.	
Postconditions: 1. New VAT configuration is saved and applied to future calculations.	
Extension Use Case: None	
Inclusion Use Case: None	
Alternative Flows: 1. InvalidVATEntered	

ID: CA-22	Use Case: Update Retail Mark-up Rate
Brief Description: Allows the Pharmacist to configure markup rates that are applied to products and transactions.	
Primary Actors: Pharmacist	
Secondary Actors: None	
Preconditions: 1. System is fully functional. 2. Pharmacist is logged into the system using their credentials. 3. Pharmacist is aware of the mark-up rates to configure on products and transactions.	
Postconditions: 1. New markup rates are saved and applied to future calculations.	
Extension Use Case: None	
Inclusion Use Case: None	
Alternative Flows: 1. InvalidRateEntered	

ID: CA-23	Use Case: Order More Stock
Brief Description: Allows Pharmacist to place an order for additional stock	
Primary Actors: Pharmacist	
Secondary Actors: IPOS-SA	
Preconditions: 1. Systems are fully functional. 2. Pharmacist is logged into the system using their credentials. 3. Pharmacist is aware of the product which needs reordering.	
Postconditions: 1. Order is submitted to the IPOS-SA subsystem.	
Extension Use Case: None	
Inclusion Use Case: None	
Alternative Flows: 1. OrderSubmissionFails	

ID: CA-24	Use Case: Generate Report
Brief Description: Generates a report, is a generalized class in which the specialised will use.	
Primary Actors: Manager	
Secondary Actors: None	
Preconditions: 1. System is fully functional. 2. Manager is logged into the system.	
Postconditions: 1. The requested report is generated for the specified period. 2. The report is displayed on screen.	
Extension Use Case: 1. Optional Printout	
Inclusion Use Case: None	
Alternative Flows: 1. NoDataAvailableForPeriod	

ID: CA-25	Use Case: Optional Printout
Brief Description: After a report is generated, it gives the manager the option to create a printout.	
Primary Actors: Manager	
Secondary Actors: None	
Preconditions: 1. A report has already been generated and displayed.	
Postconditions: 2. Report is generated successfully.	
Extension Use Case: None	
Inclusion Use Case: None	
Alternative Flows: 1. Printer used to create printout fails.	

ID: CA-26	Use Case: Generate Turnover Report
Brief Description: Generates a report showing sales turnover for a selected period.	
Primary Actors: Manager	
Secondary Actors: None	
Preconditions: Inherits preconditions from Generate Report.	
Postconditions: 1. Turnover report is generated successfully.	
Extension Use Case: Inherits extension Optional Printout	
Inclusion Use Case: None	
Specialisation of: Generate Report	
Alternative Flows: 2. NoDataAvailableForPeriod	

ID: CA-27	Use Case: Generate Stock Availability Report
------------------	---

Brief Description:	
Generates a report of current stock levels.	
Primary Actors:	
Manager	
Secondary Actors:	
None	
Preconditions:	
Inherits preconditions from Generate Report.	
Postconditions:	
1. Stock availability report is generated successfully.	
Extension Use Case:	
Inherits extension Optional Printout	
Inclusion Use Case:	
None	
Specialisation of: Generate Report	
Alternative Flows:	
1. StockLevelError	

ID: CA-28	Use Case: Generate Account Holder Debt Report
Brief Description:	

Generates a report of outstanding account holder debts.
Primary Actors: Manager
Secondary Actors: None
Preconditions: Inherits preconditions from Generate Report.
Postconditions: 1. Account debt report is generated successfully.
Extension Use Case: Inherits extension Optional Printout
Inclusion Use Case: None
Specialisation of: Generate Report
Alternative Flows: 1. NoAccountHolderWithDebt

ID: CA-29	Use Case: Place Order with IPOS-SA
Brief Description: Allows an order to be placed with the supplier's subsystem, which is IPOS-SA.	

Primary Actors: Pharmacist
Secondary Actors: IPOS-SA
Preconditions: 1. Systems are fully functional. 2. Pharmacist is logged into the system using their credentials. 3. Pharmacist is aware of which order of products need to be placed.
Postconditions: 1. Order is registered and placed with IPOS-SA.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. SupplierSystemUnavailable

ID: CA-30	Use Case: Track Order Status
Brief Description: Allows tracking the status of an order placed with IPOS-SA.	
Primary Actors:	

Pharmacist
Secondary Actors: IPOS-SA
Preconditions: 1. Systems are fully functional. 2. Pharmacist is logged into the system using their credentials. 3. Order has already been placed before by the pharmacist.
Postconditions: 1. Order status is displayed.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. OrderNotFound

ID: CA-31	Use Case: View Previous Order
Brief Description: Allows viewing of previously placed orders.	
Primary Actors: Pharmacist	

Secondary Actors: IPOS-SA
Preconditions: 1. Systems are fully functional. 2. Pharmacist is logged into the system using their credentials. 3. Previous orders exist, which were made by the pharmacist.
Postconditions: 1. List of previous orders is displayed.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. NoPreviousOrdersMade

ID: CA-32	Use Case: Query Outstanding Balance
Brief Description: Allows querying an account holder's outstanding balance.	
Primary Actors: Pharmacist	

Secondary Actors: None
Preconditions: 1. System is fully functional. 2. Pharmacist is logged into the system using their credentials. 3. Account holder exists who still yet has an outstanding balance (which the pharmacist can see).
Postconditions: 1. Outstanding balance is displayed to the pharmacist.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. AccountHolderNotFound

ID: CA-33	Use Case: Set Credit Limit
Brief Description: Allows the updating/setting of a credit limit for an account holder.	
Primary Actors: Manager	
Secondary Actors:	

None
Preconditions: 1. System is fully functional. 2. Manager is logged into the system using their credentials. 3. Account holder exists. 4. Manager is aware of which account holder requires a credit limit set.
Postconditions: 1. Credit limit is updated/set.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. InvalidCreditLimitEntered

ID: CA-34	Use Case: Apply Discount Plan
Brief Description: Applies a discount plan to an account holder.	
Primary Actors: Pharmacist	
Secondary Actors: None	

Preconditions: 1. System is fully functional. 2. Pharmacist is logged into the system using their credentials. 3. Account holder exists. 4. Pharmacist is aware which account holder requires a discount plan to be applied.
Postconditions: 1. Discount plan is applied to the account holder.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. InvalidDiscountPlanSelected

ID: CA-35	Use Case: Update Reminder Template
Brief Description: Allows the updating of the template used for payment reminders.	
Primary Actors: Manager	
Secondary Actors: None	
Preconditions:	

1. System is fully functional. 2. Manager is logged into the system using their credentials. 3. Manager is aware which payment reminder template needs to be updated.
Postconditions: 1. Reminder template is updated.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. InvalidPaymentTemplateFormat

ID: CA-36	Use Case: Update Receipt/Invoice Template
Brief Description: Allows the updating of the receipt template.	
Primary Actors: Manager	
Secondary Actors: None	
Preconditions: 1. System is fully functional. 2. Manager is logged into the system using their credentials.	

3. Manager is aware which receipt template needs to be updated.
Postconditions: 1. Receipt template is updated.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. InvalidReceiptTemplateFormat

ID: CA-37	Use Case: Update Merchant Identity Details
Brief Description: Allows updating the pharmacy's identity details used on receipts and documents.	
Primary Actors: Manager	
Secondary Actors: None	
Preconditions: 1. System is fully functional. 2. Manager is logged into the system using their credentials. 3. Manager is aware which merchant's identity details need to be updated.	

Postconditions: 1. Merchant identity details are updated.
Extension Use Case: None
Inclusion Use Case: None
Alternative Flows: 1. InvalidDetailsProvided

5.2.1.3 Prioritised List of use cases